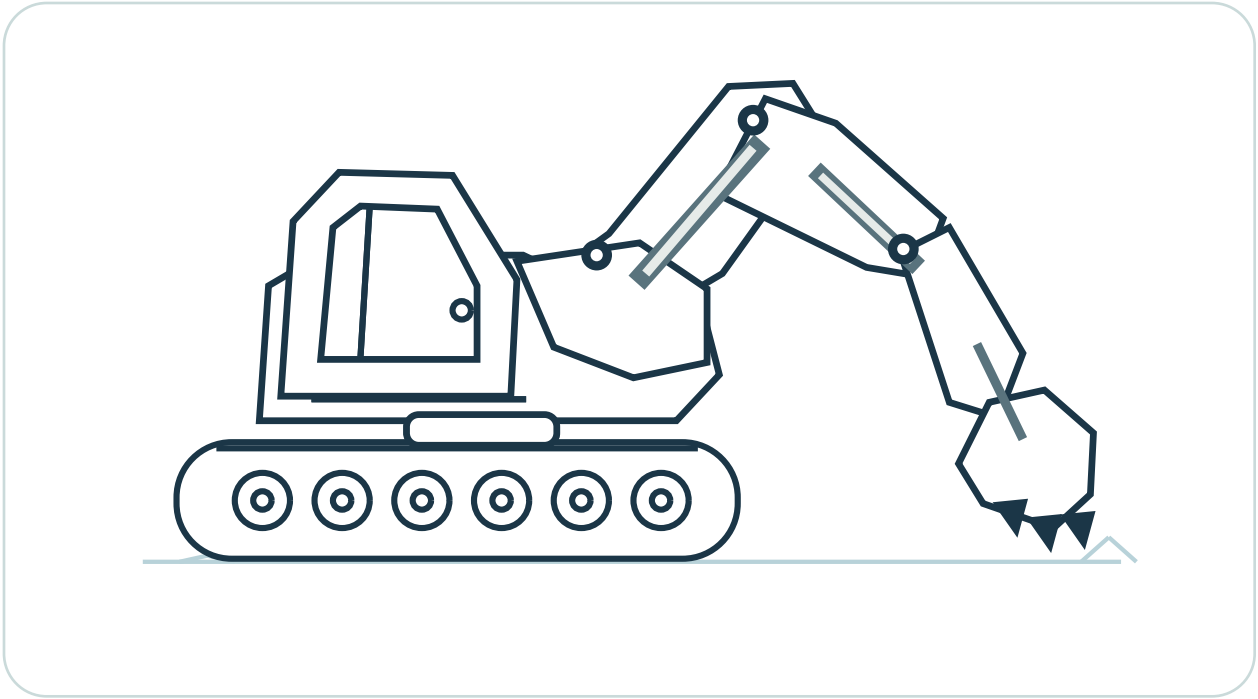
HOW TO DRAW CONSTRUCTION MACHINES

Step-by-Step Trucks, Diggers, Cranes, and More

for Kids Ages 6-10



30 PROJECTS / 10 SCENE CHALLENGES

Heath Tolley

WELCOME

Start with shapes

01 Look

See the whole machine before you draw.

02 Build

Copy the shapes in order. New parts appear each step.

03 Check

Compare the final drawing with your own version.

04 Invent

Finish the challenge with a useful scene.

You need a pencil, an eraser, and a sheet of paper.

Try light lines first. Trace darker lines after the machine looks right.

Your drawing can be different. That is part of learning.

Erase hidden lines when a new part covers an earlier shape.

Final draft | Revision 15 | 30 September 2026

WARM-UP

Your shape toolkit



BOX

truck bodies and cabins



CIRCLE

wheels and drums



LONG LINE

booms and blades



ZIGZAG

tracks, treads, and lifting arms

Practice each shape five times, then combine a box and two circles.

MAP

How the lessons work

Level 1 / Shape builders

Projects 1-10: recognize the body and wheels.

Level 2 / Long machines

Projects 11-20: add arms, booms, and special parts.

Level 3 / Clever details

Projects 21-30: compare similar machines.

Finish / Scene challenges

Make whole construction scenes with a purpose.

Tip: Go back to an easier project whenever you need a warm-up.

LEVEL 1 / PROJECT 01

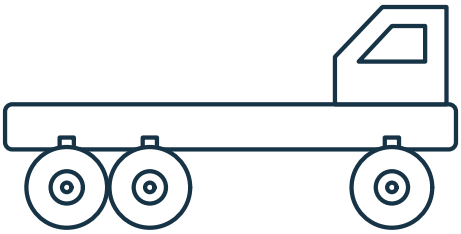
Dump truck

A dump bed tips to unload gravel.

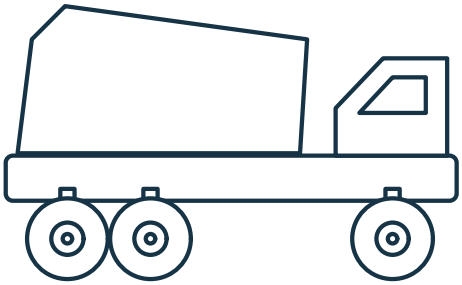
1 Draw the truck frame



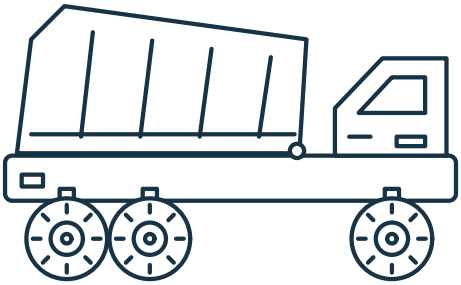
2 Add cab and tires



3 Build the tipping bed



4 Add ribs and hinge



TRY IT

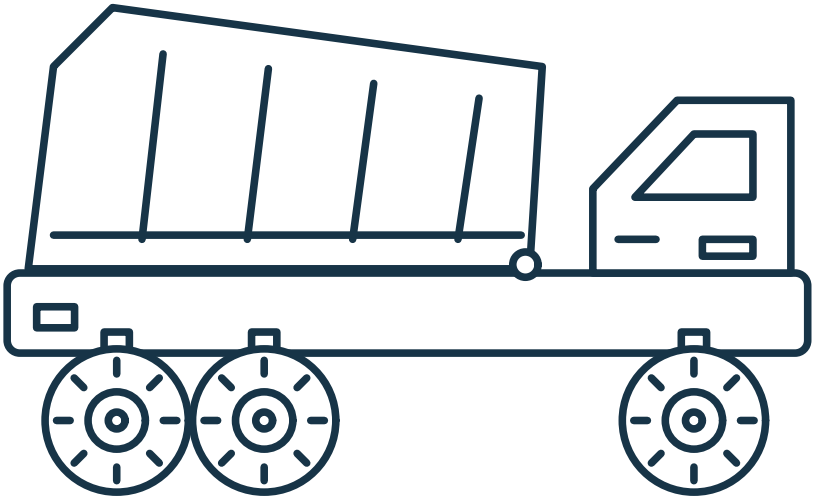
Add gravel falling from the raised bed.

Draw on your own paper. Change one detail to make it yours.

PROJECT 01 / PRACTICE

Dump truck: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Add gravel falling from the raised bed.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 02

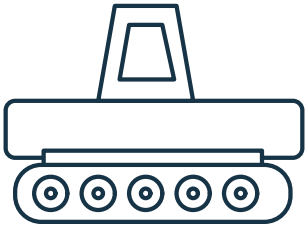
Bulldozer

A wide blade pushes soil across a site.

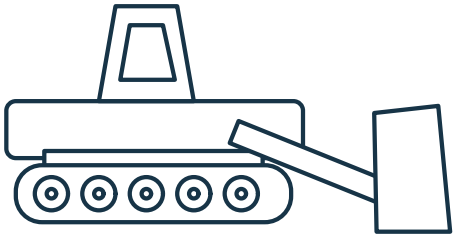
1 Draw the crawler body



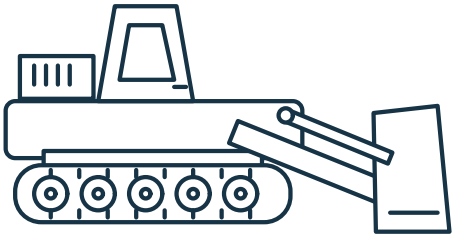
2 Add tracks and cab



3 Attach the pushing blade



4 Add cylinder and engine



TRY IT

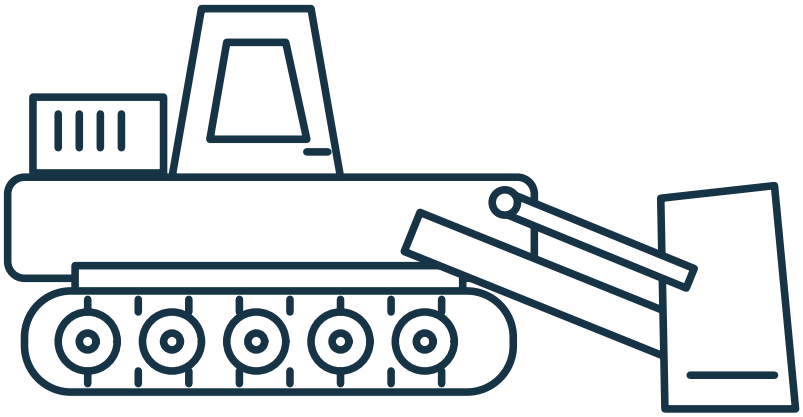
Show a mound of dirt in front of the blade.

Draw on your own paper. Change one detail to make it yours.

PROJECT 02 / PRACTICE

Bulldozer: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Show a mound of dirt in front of the blade.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 03

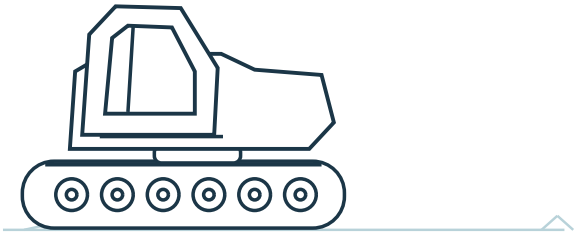
Excavator

A long arm digs with a bucket.

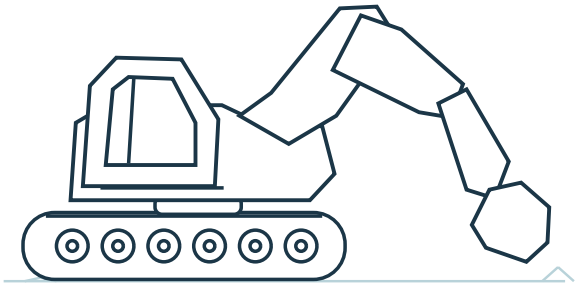
1 Draw tracks and rollers



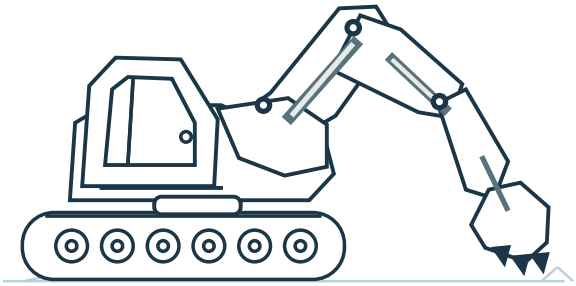
2 Add body and glass cab



3 Build the digging arm



4 Add pistons and teeth



TRY IT

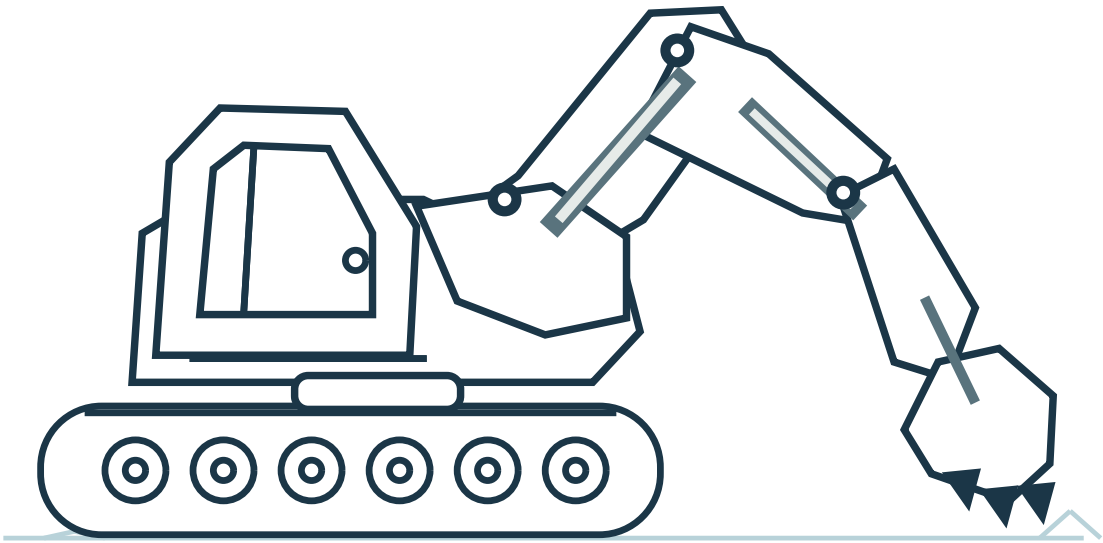
Draw a trench below the bucket.

Draw on your own paper. Change one detail to make it yours.

PROJECT 03 / PRACTICE

Excavator: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw a trench below the bucket.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 04

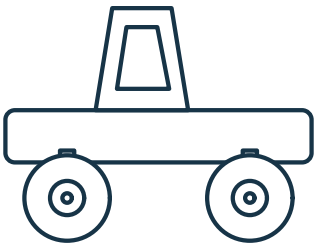
Wheel loader

A front bucket scoops loose material.

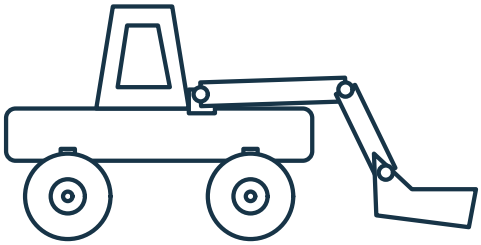
1 Draw the loader frame



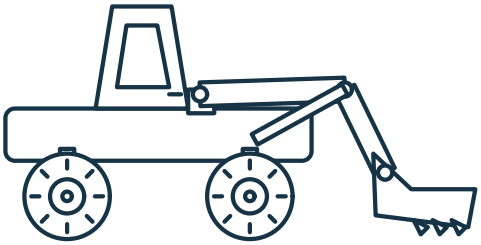
2 Add tires and cab



3 Join lift arms and bucket



4 Add cylinder and teeth



TRY IT

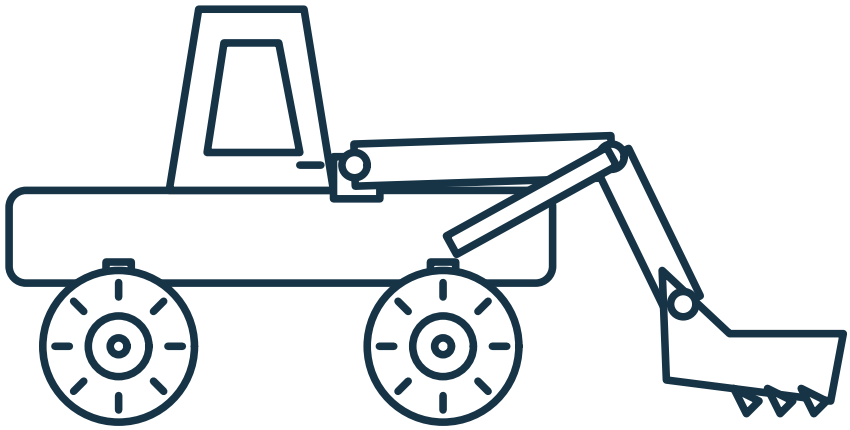
Fill the bucket with stones.

Draw on your own paper. Change one detail to make it yours.

PROJECT 04 / PRACTICE

Wheel loader: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Fill the bucket with stones.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 05

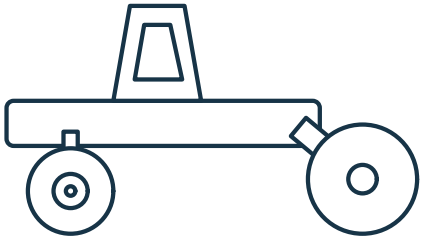
Road roller

Heavy drums flatten a new road.

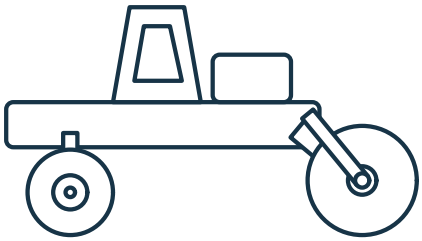
1 Draw the frame



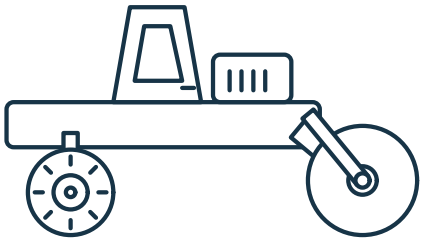
2 Mount tire, drum, and cab



3 Add engine and drum fork



4 Finish vents and axle



TRY IT

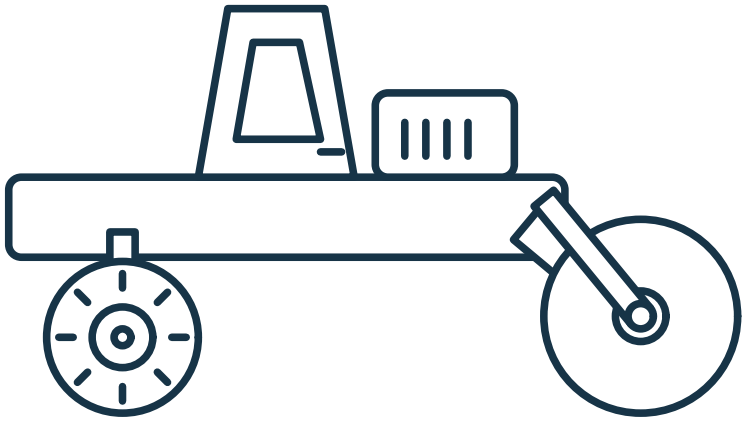
Add a smooth road behind the roller.

Draw on your own paper. Change one detail to make it yours.

PROJECT 05 / PRACTICE

Road roller: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Add a smooth road behind the roller.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 06

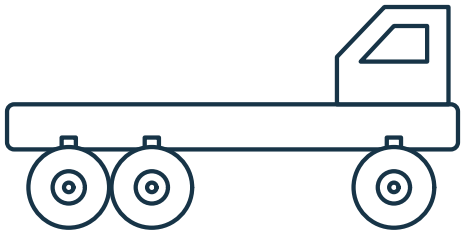
Cement mixer

A turning drum mixes concrete.

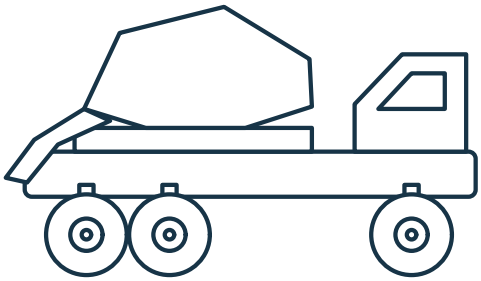
1 Draw the truck frame



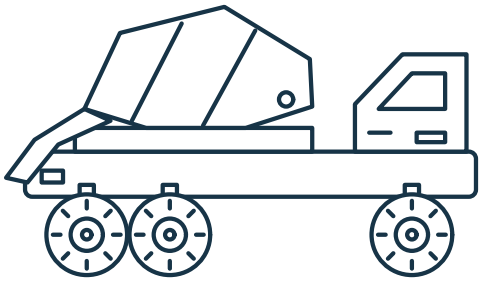
2 Add cab and tires



3 Mount drum and chute



4 Add drum bands and pins



TRY IT

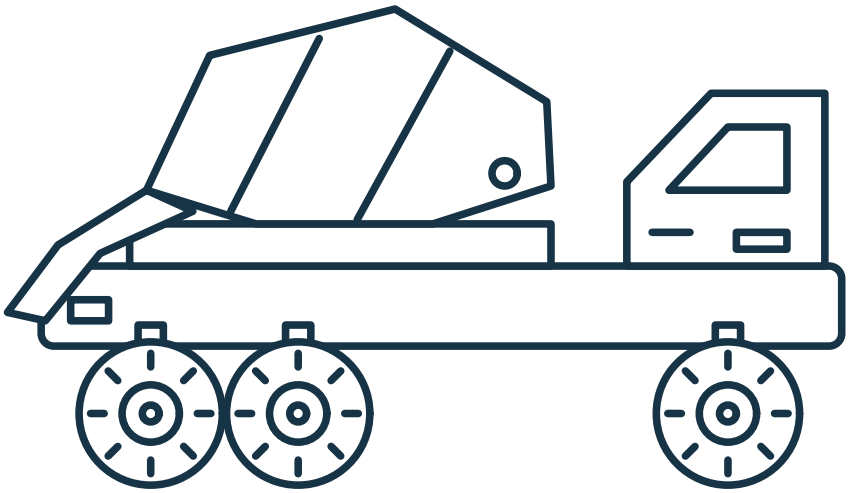
Add a chute and wet concrete.

Draw on your own paper. Change one detail to make it yours.

PROJECT 06 / PRACTICE

Cement mixer: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Add a chute and wet concrete.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 07

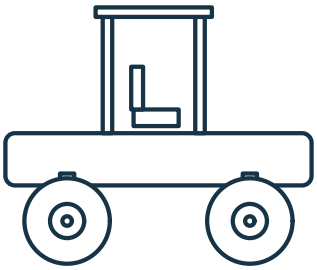
Forklift

Two metal forks lift a pallet.

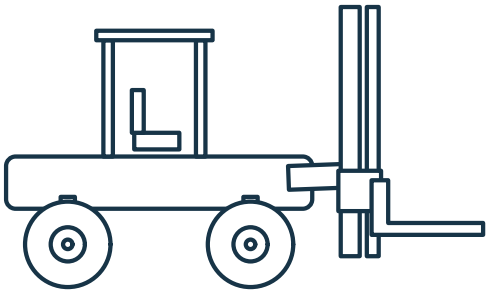
1 Draw the low body



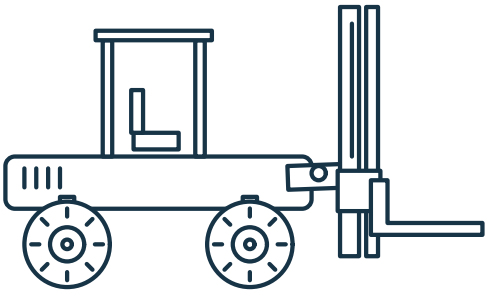
2 Add tires, seat, and guard



3 Mount mast and forks



4 Add guides and hinge



TRY IT

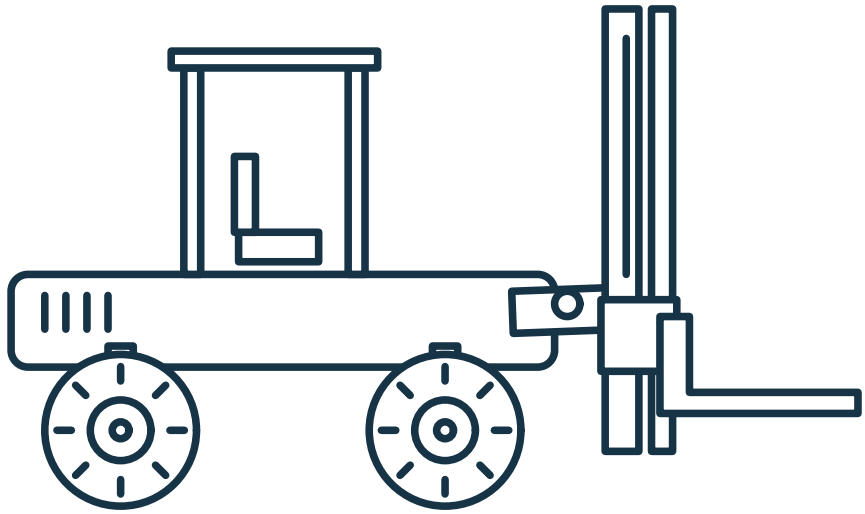
Stack three boxes on a pallet.

Draw on your own paper. Change one detail to make it yours.

PROJECT 07 / PRACTICE

Forklift: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Stack three boxes on a pallet.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 08

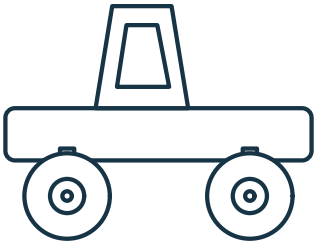
Skid steer

A compact loader turns in tight spaces.

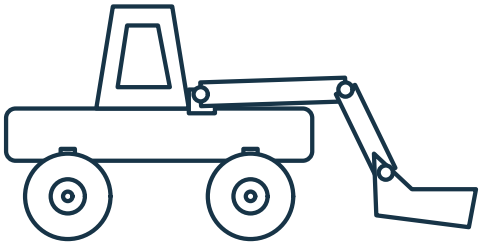
1 Draw the compact frame



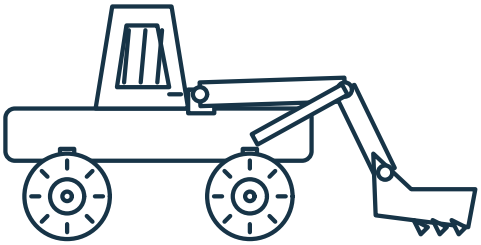
2 Add tires and cab



3 Mount arms and bucket



4 Add cylinder and cab bars



TRY IT

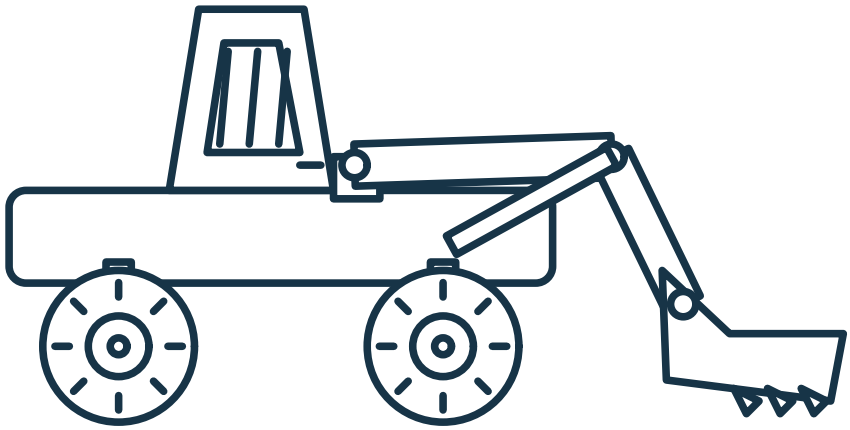
Show the machine inside a narrow gate.

Draw on your own paper. Change one detail to make it yours.

PROJECT 08 / PRACTICE

Skid steer: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Show the machine inside a narrow gate.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 09

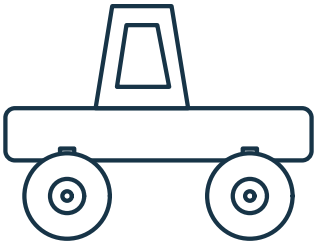
Backhoe

A backhoe digs at the rear and loads at the front.

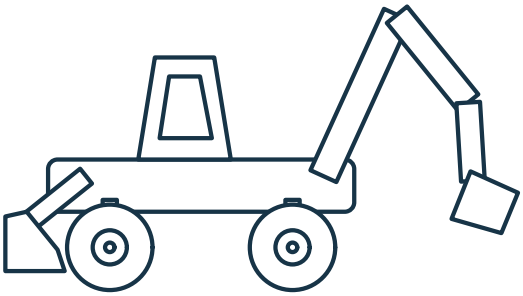
1 Draw the main frame



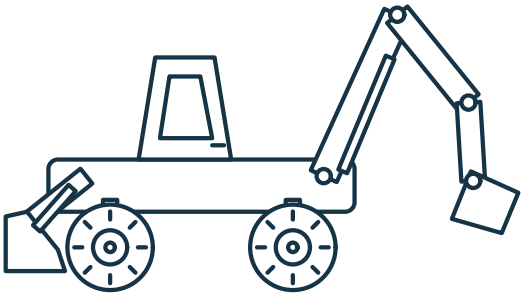
2 Add tires and cab



3 Attach both working ends



4 Add pivots and cylinders



TRY IT

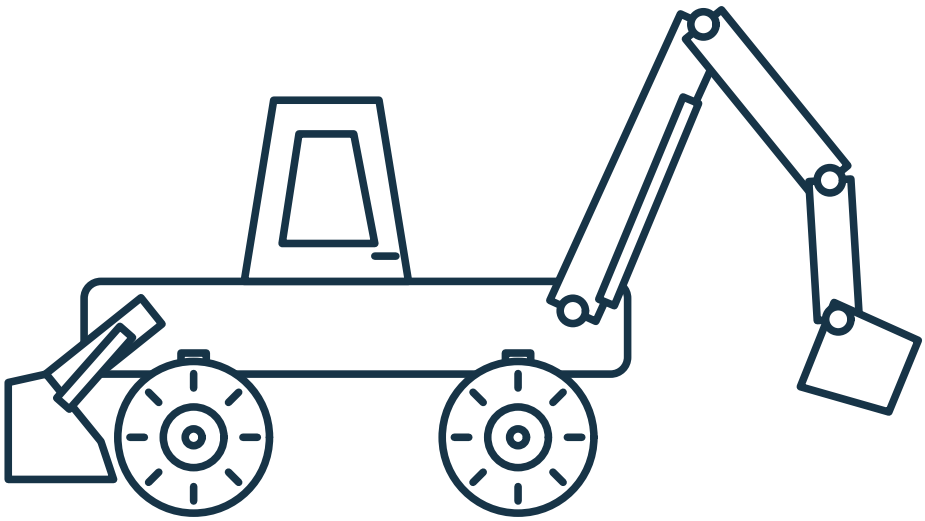
Add dirt beside both ends.

Draw on your own paper. Change one detail to make it yours.

PROJECT 09 / PRACTICE

Backhoe: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Add dirt beside both ends.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 1 / PROJECT 10

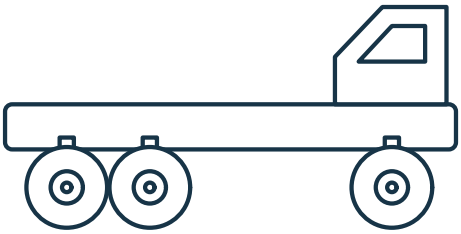
Mobile crane

A telescoping boom lifts heavy loads.

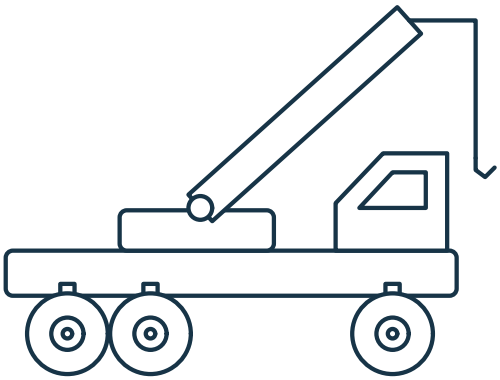
1 Draw the truck frame



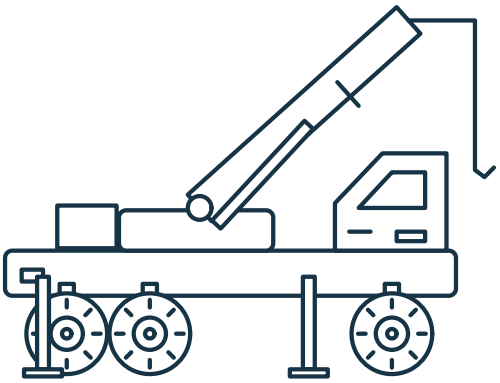
2 Add cab and tires



3 Mount boom and hook



4 Add cylinder and stabilizers



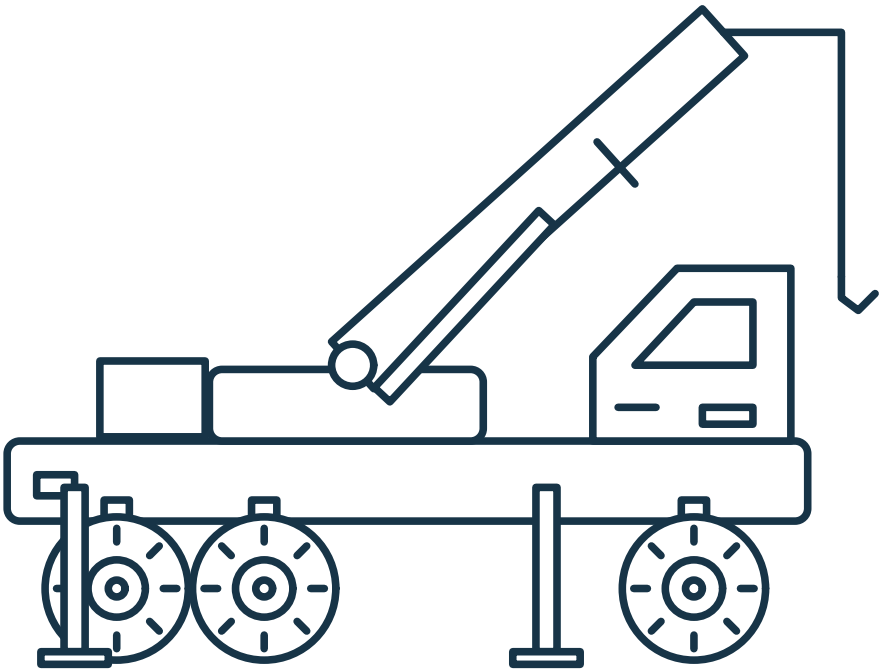
TRY IT

Hang a safe load from the hook.

Draw on your own paper. Change one detail to make it yours.

Mobile crane: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Hang a safe load from the hook.

Circle two moving parts. Add a person for scale.

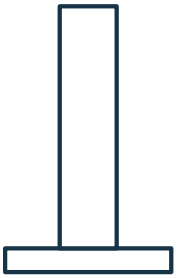
Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 11

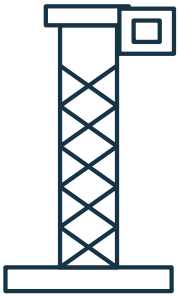
Tower crane

A tall fixed crane works above buildings.

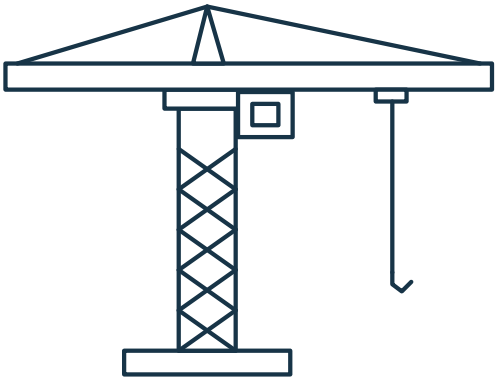
1 Draw foundation and mast



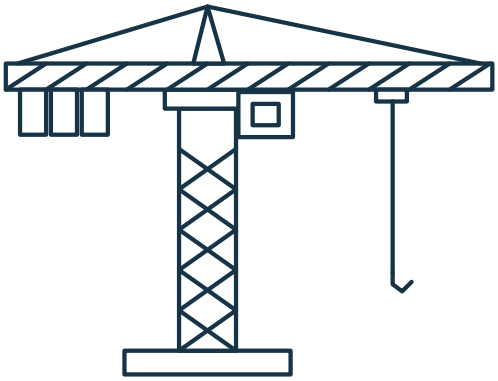
2 Add mast braces and cab



3 Build jib and hoist



4 Add weights and bracing



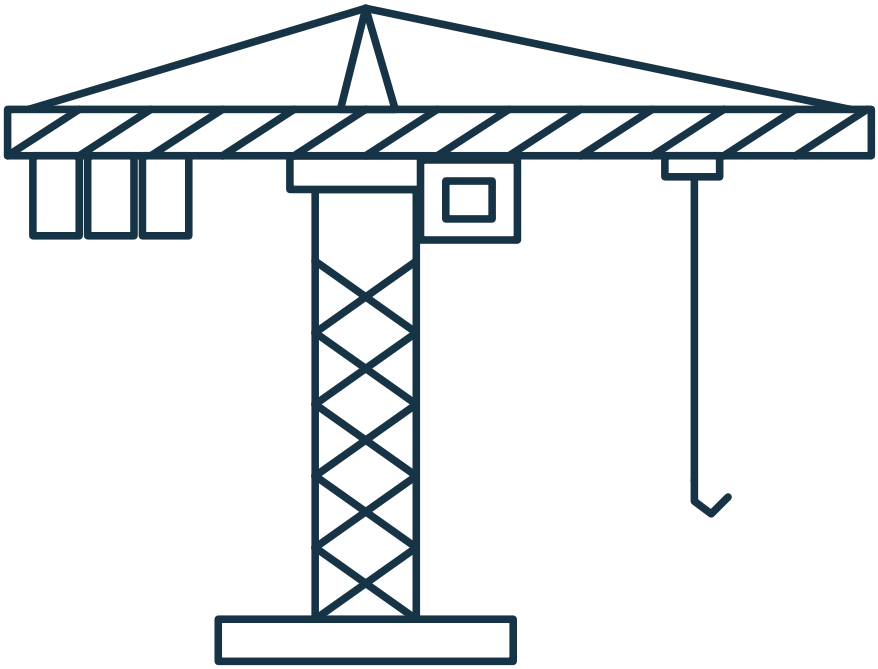
TRY IT

Draw a building beneath the hook.

Draw on your own paper. Change one detail to make it yours.

Tower crane: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw a building beneath the hook.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 12

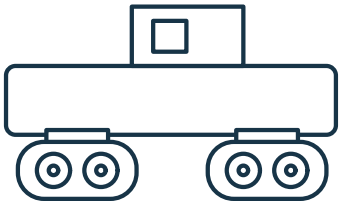
Paver

A paver spreads hot asphalt in a flat strip.

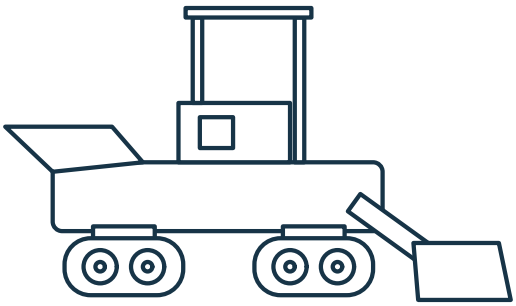
1 Draw the paver body



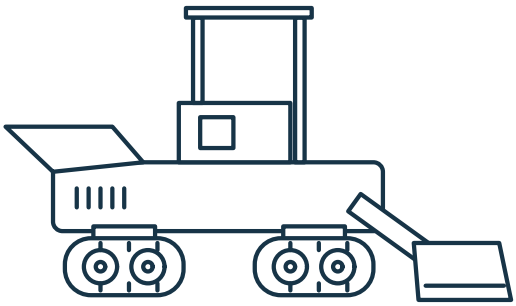
2 Mount tracks and controls



3 Add hopper and screed



4 Finish screed and vents



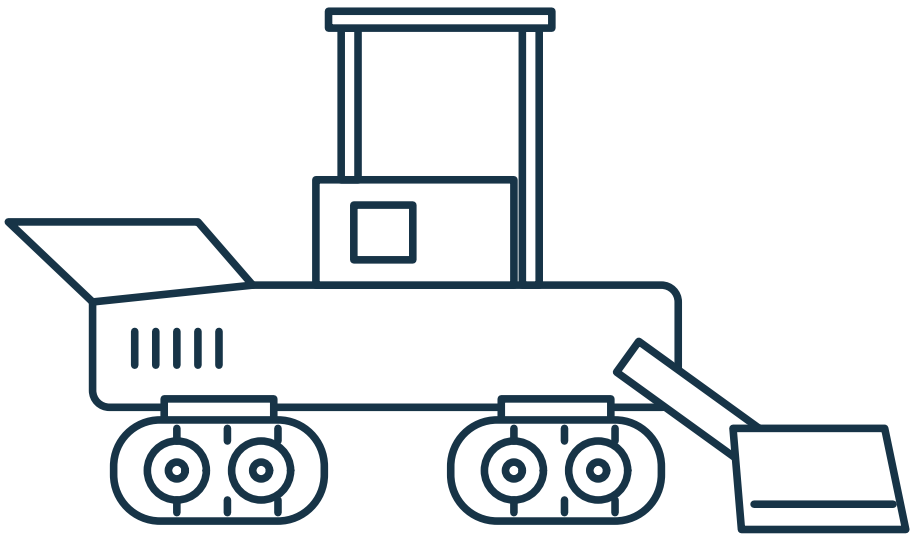
TRY IT

Show the fresh road behind it.

Draw on your own paper. Change one detail to make it yours.

Paver: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Show the fresh road behind it.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 13

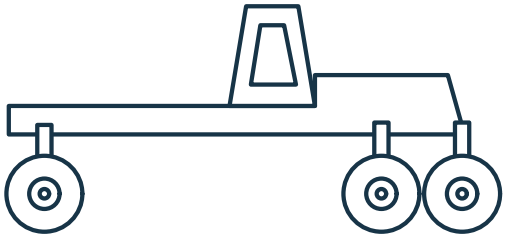
Motor grader

A center blade smooths and shapes the ground.

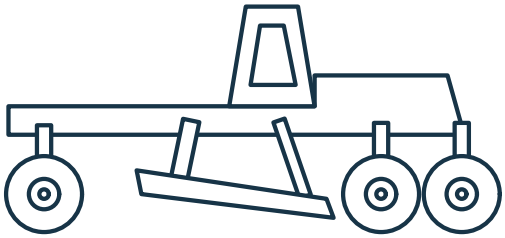
1 Draw the closed frame



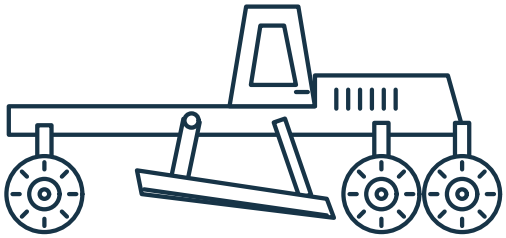
2 Mount tires and cab



3 Attach the center blade



4 Add joints and engine vents



TRY IT

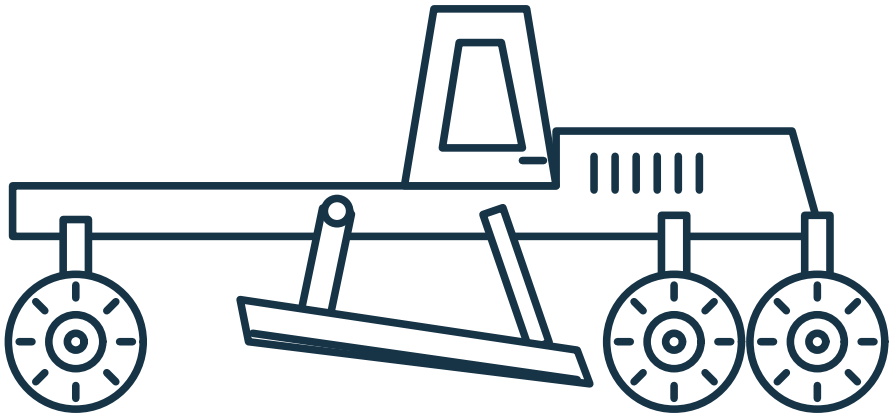
Draw a crown in the road.

Draw on your own paper. Change one detail to make it yours.

PROJECT 13 / PRACTICE

Motor grader: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw a crown in the road.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 14

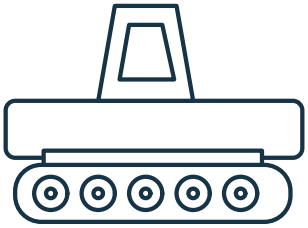
Trencher

A digging chain makes a narrow trench.

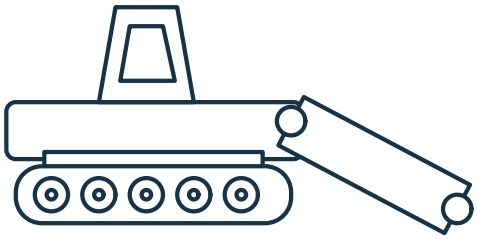
1 Draw the crawler body



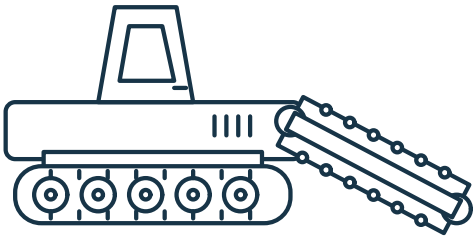
2 Mount tracks and cab



3 Attach the chain boom



4 Add chain and cutting teeth



TRY IT

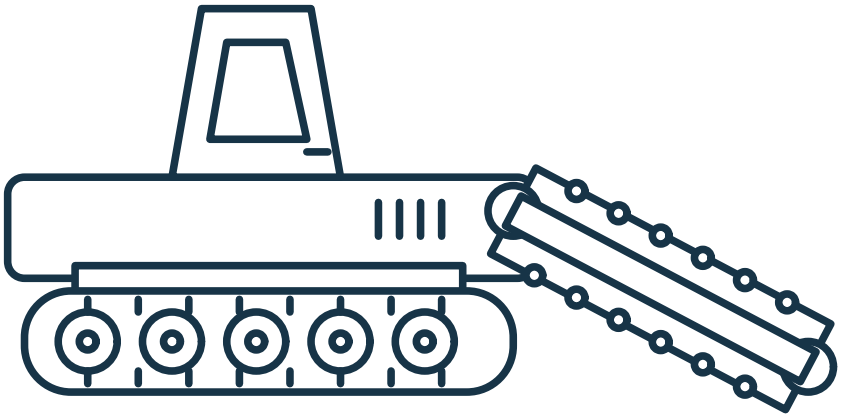
Add a pipe that could fit the trench.

Draw on your own paper. Change one detail to make it yours.

PROJECT 14 / PRACTICE

Trencher: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Add a pipe that could fit the trench.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 15

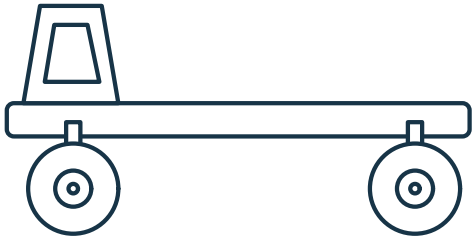
Scraper

A scraper cuts, carries, and spreads soil.

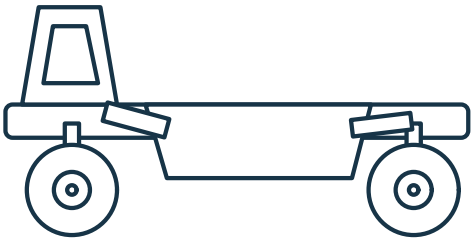
1 Draw the long frame



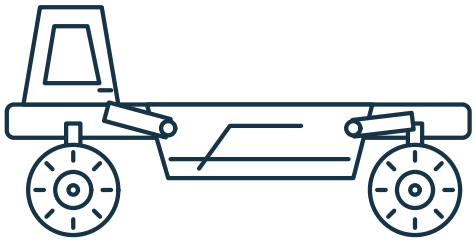
2 Mount tires and cab



3 Build the soil bowl



4 Add blade and hinges



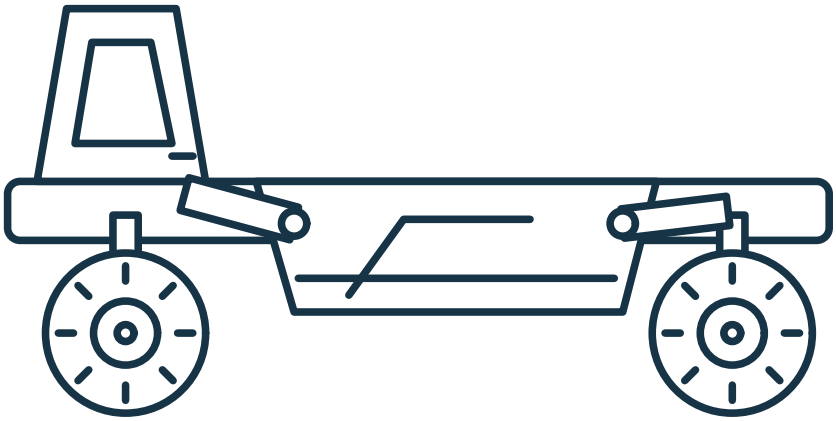
TRY IT

Shade the soil in its bowl.

Draw on your own paper. Change one detail to make it yours.

Scraper: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Shade the soil in its bowl.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 16

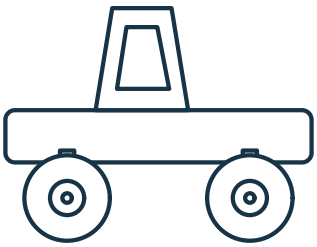
Telehandler

A long lifting boom reaches high places.

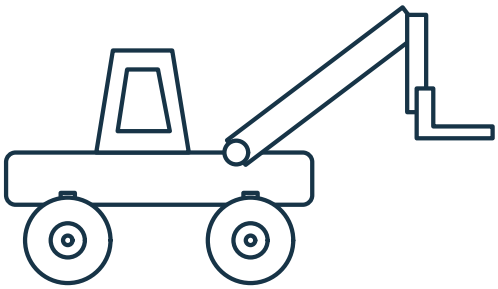
1 Draw the carrier



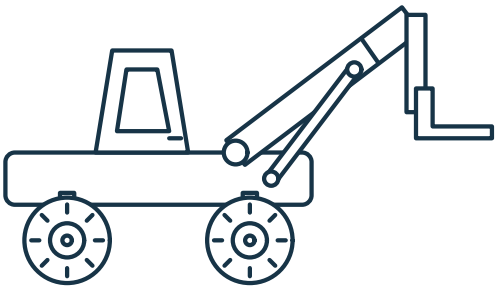
2 Mount tires and cab



3 Attach boom and forks



4 Add lift cylinder and seam



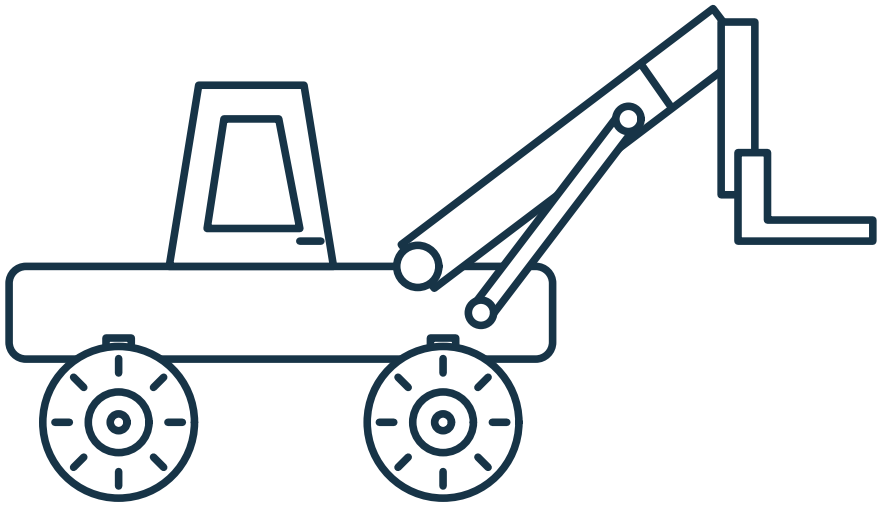
TRY IT

Put a crate on the forks.

Draw on your own paper. Change one detail to make it yours.

Telehandler: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Put a crate on the forks.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 17

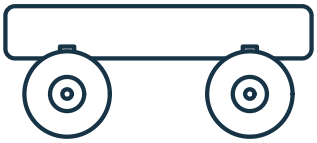
Scissor lift

Crossed supports raise a work platform.

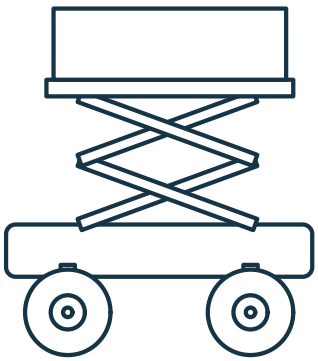
1 Draw the low base



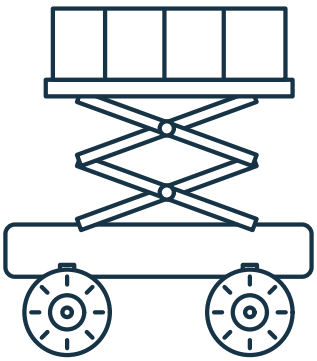
2 Mount the wheels



3 Build scissors and platform



4 Add pins and guardrails



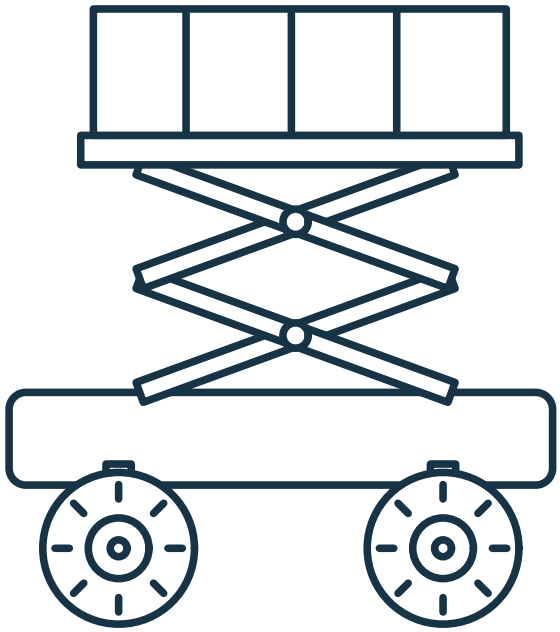
TRY IT

Draw a worker wearing a hard hat.

Draw on your own paper. Change one detail to make it yours.

Scissor lift: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw a worker wearing a hard hat.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 18

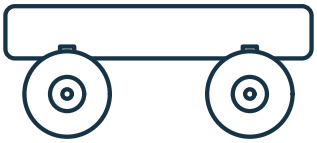
Boom lift

A jointed arm raises a small basket.

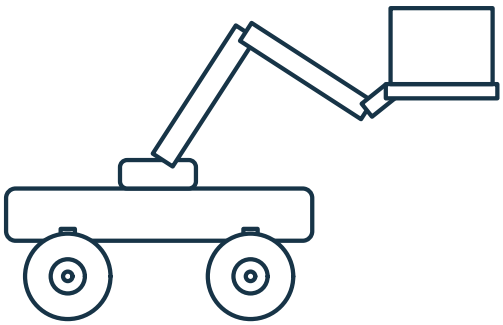
1 Draw the low base



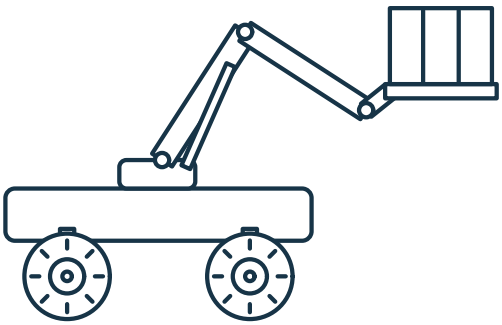
2 Mount the wheels



3 Join arms and work basket



4 Add pivots and basket rails



TRY IT

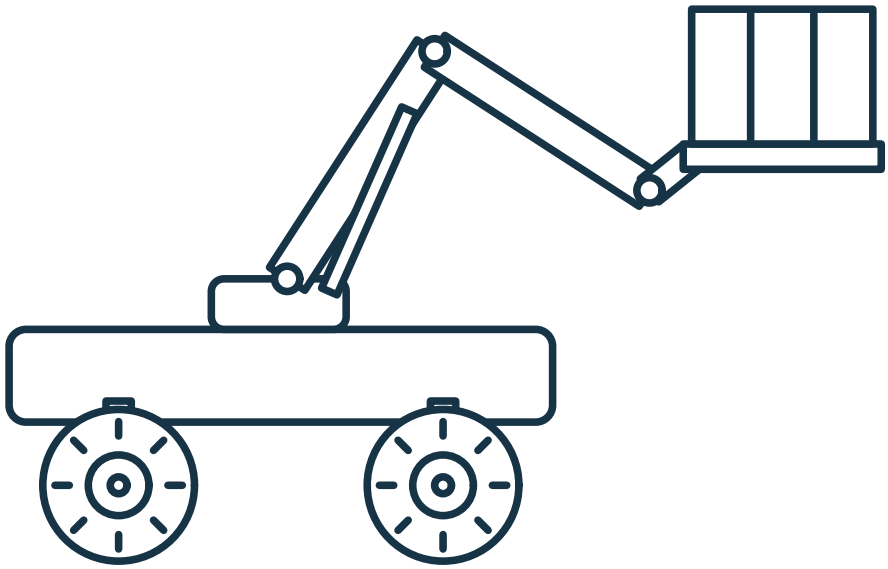
Show a light being repaired.

Draw on your own paper. Change one detail to make it yours.

PROJECT 18 / PRACTICE

Boom lift: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Show a light being repaired.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 19

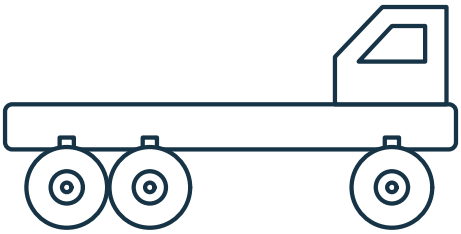
Concrete pump

A folding boom sends concrete through a hose.

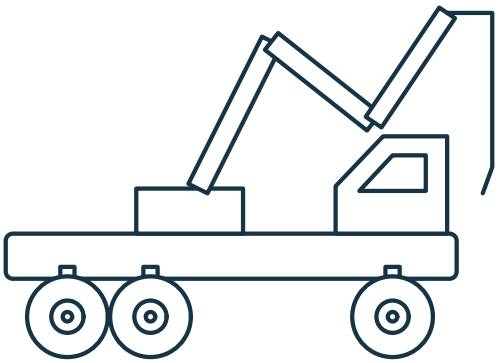
1 Draw the truck frame



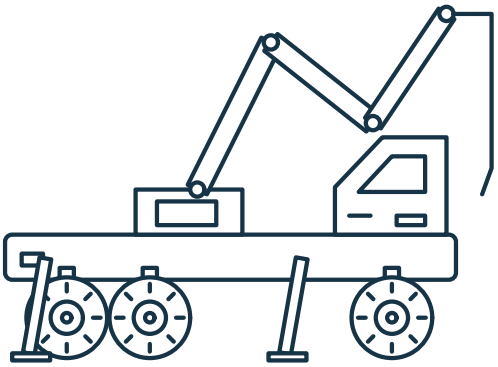
2 Add cab and tires



3 Join boom and delivery hose



4 Add joints and stabilizers



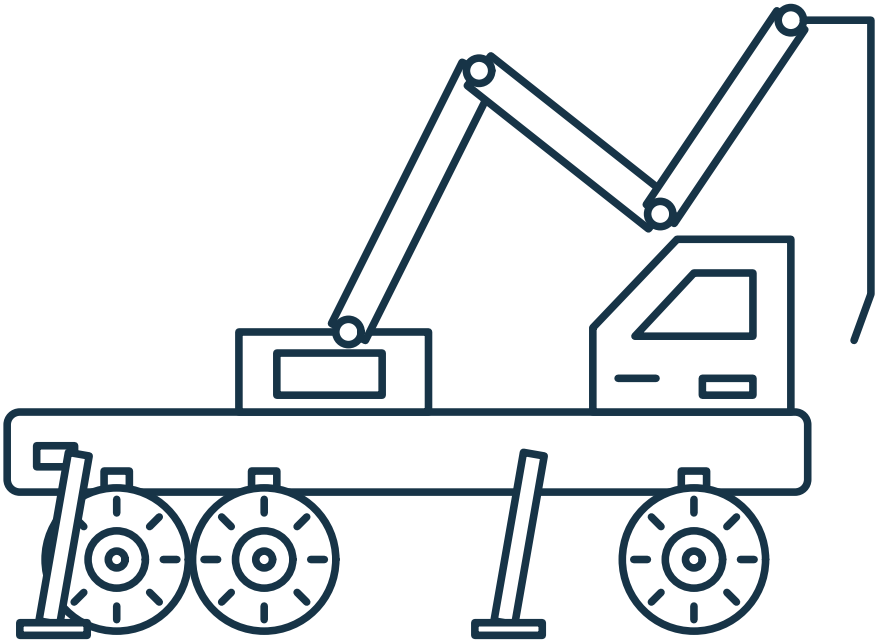
TRY IT

Draw the foundation being poured.

Draw on your own paper. Change one detail to make it yours.

Concrete pump: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw the foundation being poured.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 2 / PROJECT 20

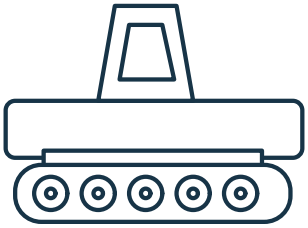
Drilling rig

A rotating drill makes deep holes.

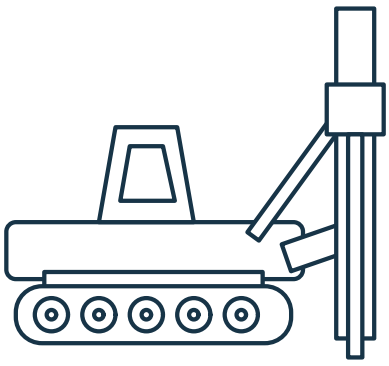
1 Draw the crawler body



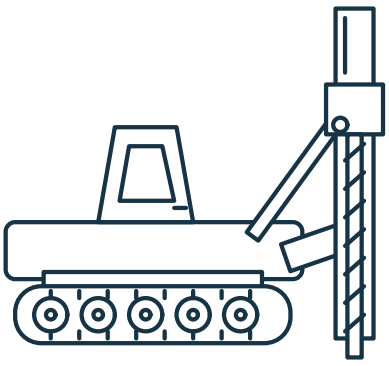
2 Mount tracks and cab



3 Connect mast and drill



4 Add guide and spiral flutes



TRY IT

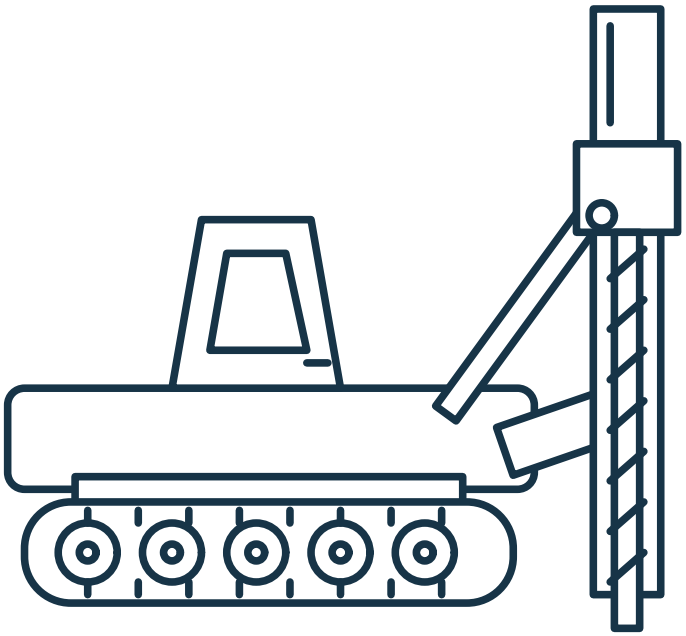
Extend the drill below the ground line.

Draw on your own paper. Change one detail to make it yours.

PROJECT 20 / PRACTICE

Drilling rig: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Extend the drill below the ground line.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 21

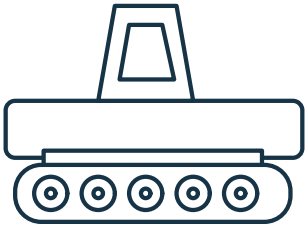
Pile driver

A tall mast guides a heavy hammer.

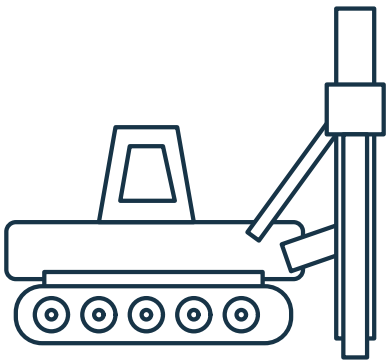
1 Draw the crawler body



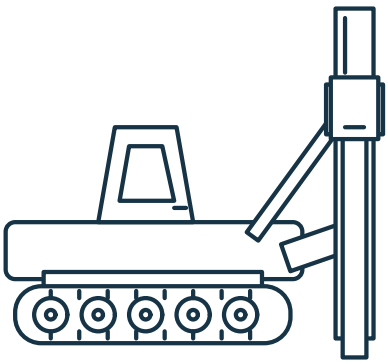
2 Mount tracks and cab



3 Connect mast and pile



4 Add hammer and guide



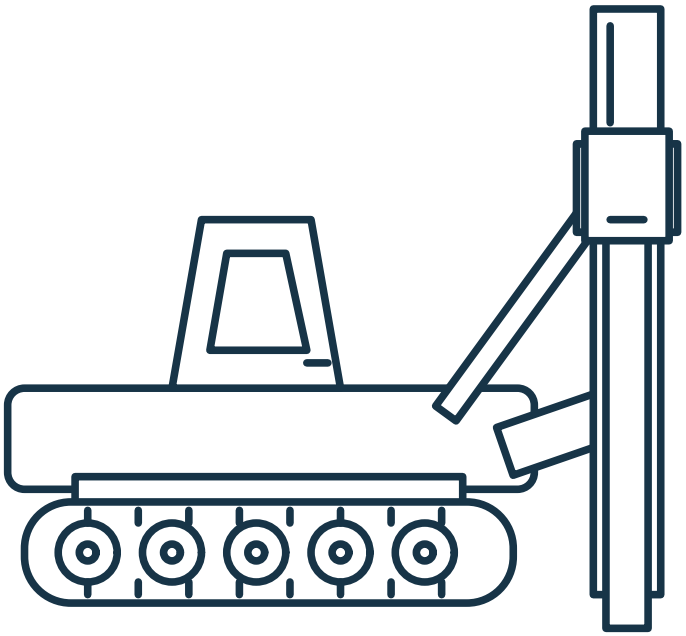
TRY IT

Draw one pile entering the earth.

Draw on your own paper. Change one detail to make it yours.

Pile driver: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw one pile entering the earth.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 22

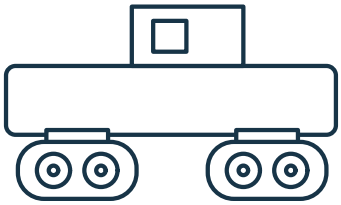
Asphalt mill

A milling drum removes old pavement.

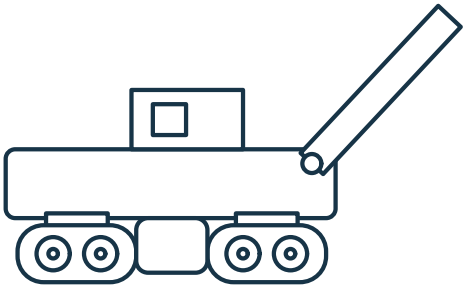
1 Draw the machine body



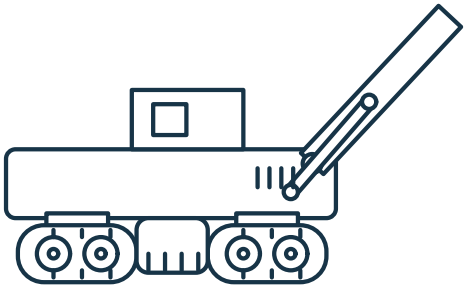
2 Mount tracks and controls



3 Attach drum and conveyor



4 Add support and cutter teeth



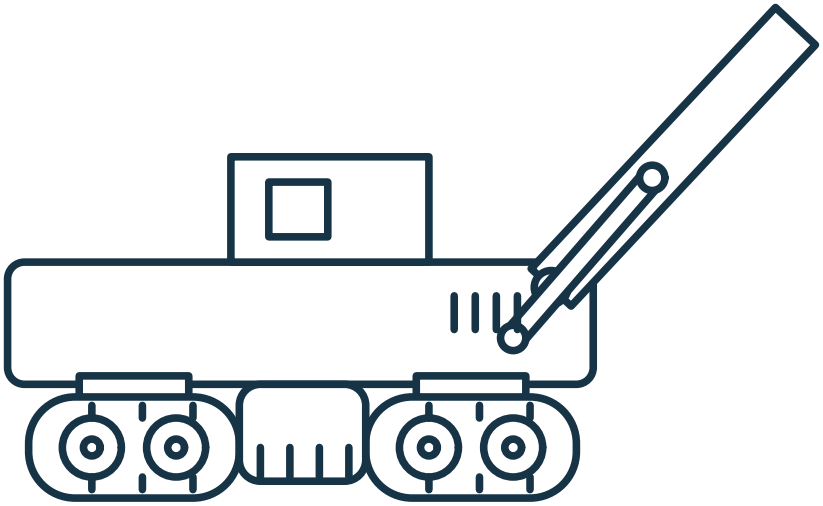
TRY IT

Show broken pavement behind it.

Draw on your own paper. Change one detail to make it yours.

Asphalt mill: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Show broken pavement behind it.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 23

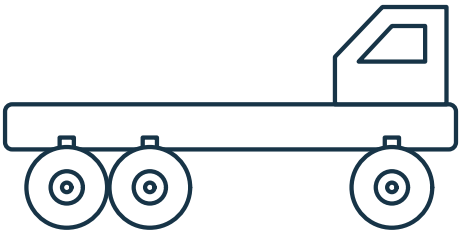
Snowplow

An angled blade clears snow from roads.

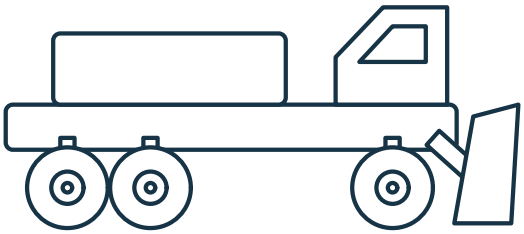
1 Draw the truck frame



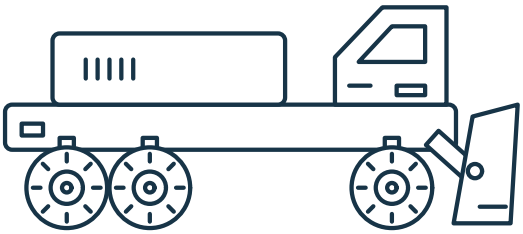
2 Add cab and tires



3 Mount body and plow blade



4 Add hinge and blade edge



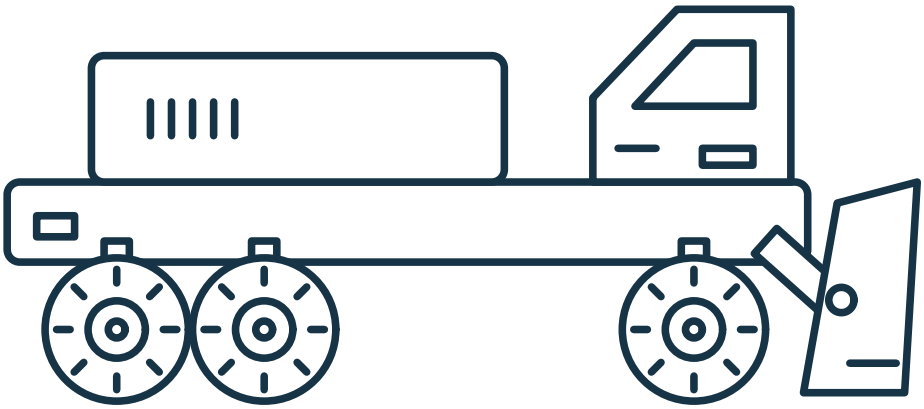
TRY IT

Show cleared and snowy lanes.

Draw on your own paper. Change one detail to make it yours.

Snowplow: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Show cleared and snowy lanes.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 24

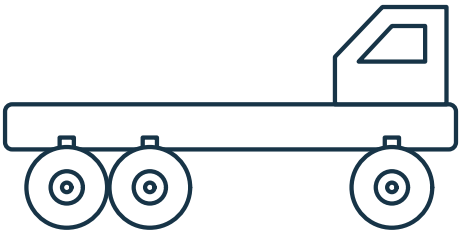
Street sweeper

Brushes collect dust and debris.

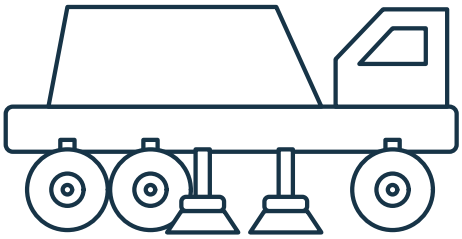
1 Draw the truck frame



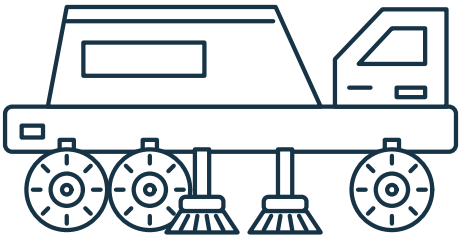
2 Add cab and tires



3 Mount hopper and brushes



4 Add bristles and service hatch



TRY IT

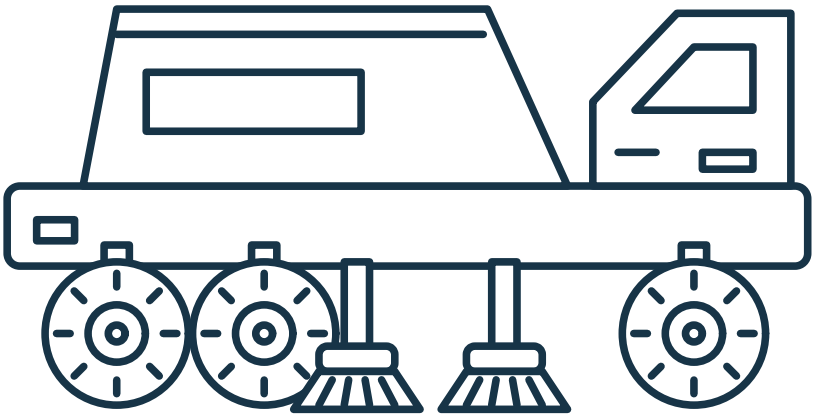
Add leaves in front of the brushes.

Draw on your own paper. Change one detail to make it yours.

PROJECT 24 / PRACTICE

Street sweeper: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Add leaves in front of the brushes.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 25

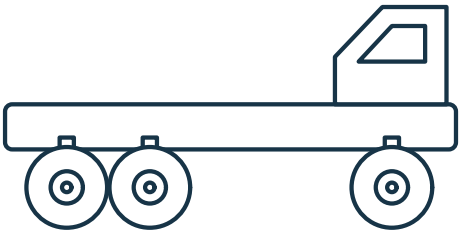
Tow truck

A lifting arm moves a disabled vehicle.

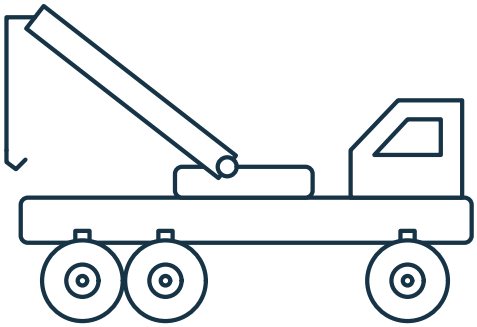
1 Draw the truck frame



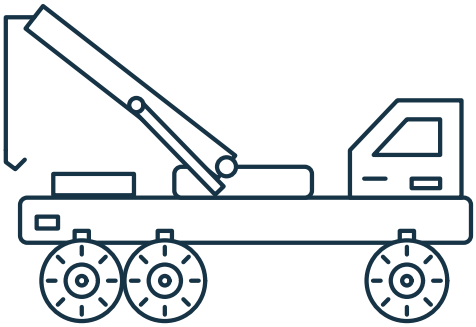
2 Add cab and tires



3 Mount rear boom and hook



4 Add cylinder and toolbox



TRY IT

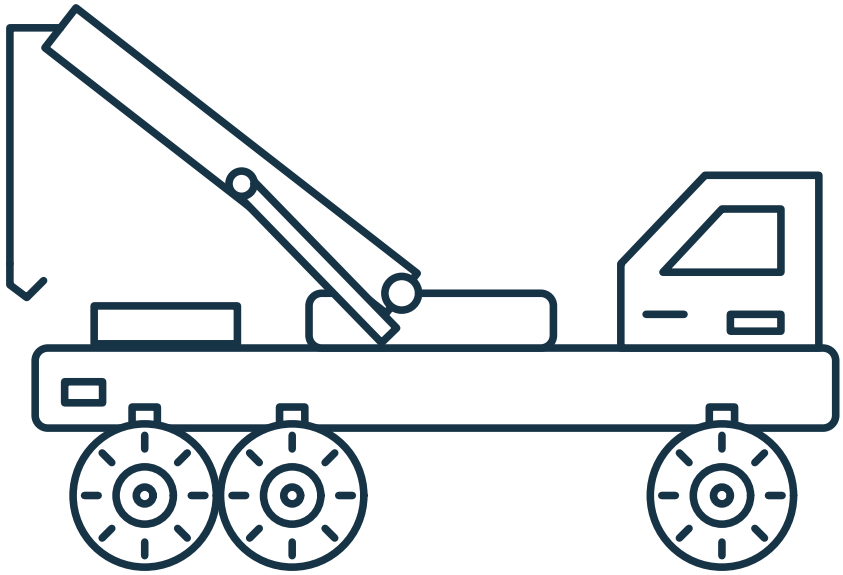
Draw a small car behind it.

Draw on your own paper. Change one detail to make it yours.

PROJECT 25 / PRACTICE

Tow truck: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw a small car behind it.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 26

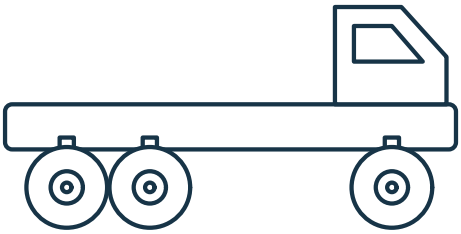
Fire engine

Firefighters carry hoses and ladders.

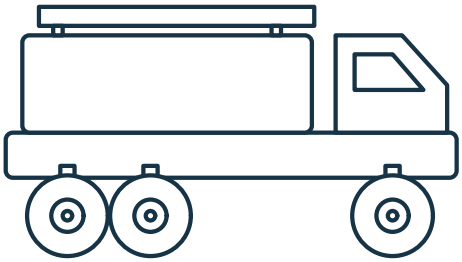
1 Draw the truck frame



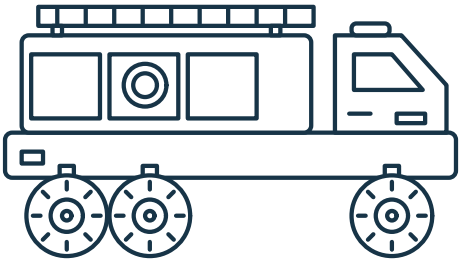
2 Add cab and tires



3 Build body and ladder



4 Add hose reel and lockers



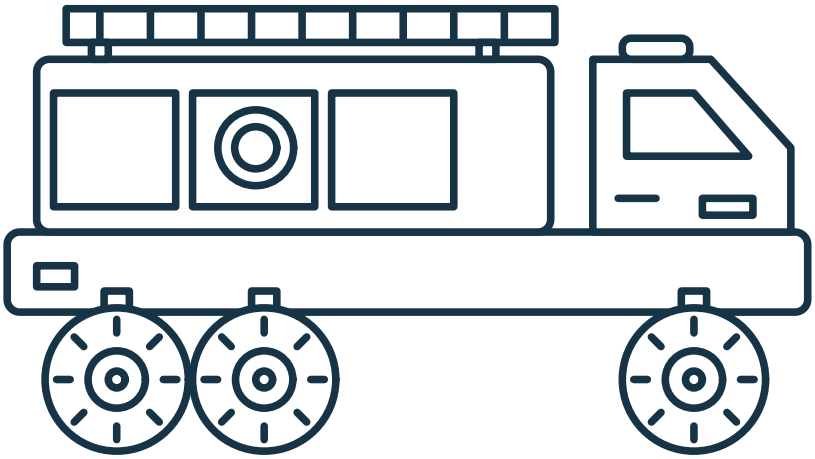
TRY IT

Add a ladder and a hydrant.

Draw on your own paper. Change one detail to make it yours.

Fire engine: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Add a ladder and a hydrant.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 27

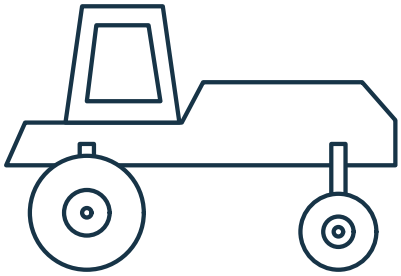
Farm tractor

Large rear tires give a tractor traction.

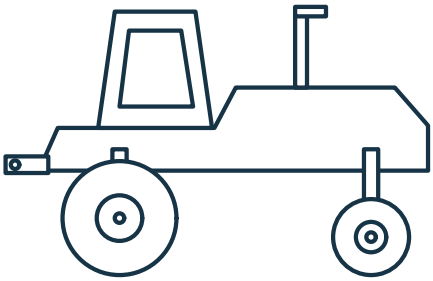
1 Shape frame and hood



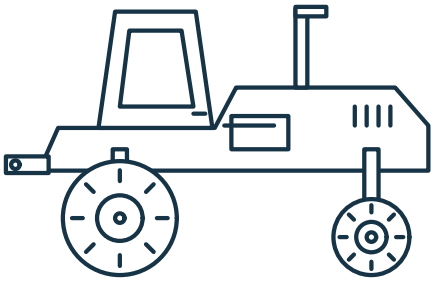
2 Mount cab and both tires



3 Add exhaust and rear hitch



4 Add grille and access panel



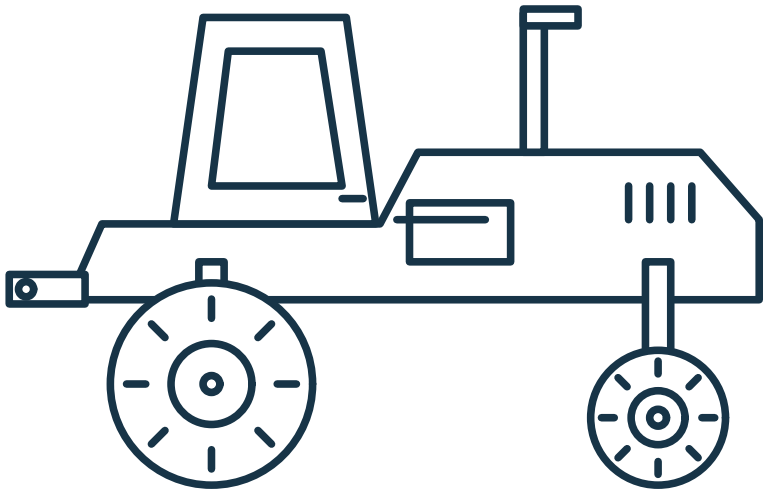
TRY IT

Attach a wagon behind the tractor.

Draw on your own paper. Change one detail to make it yours.

Farm tractor: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Attach a wagon behind the tractor.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 28

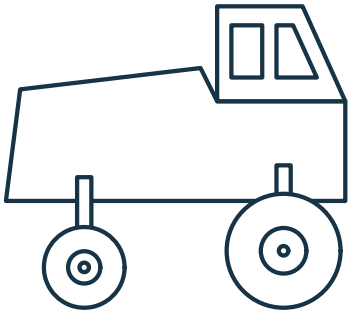
Combine harvester

A wide header cuts grain in a field.

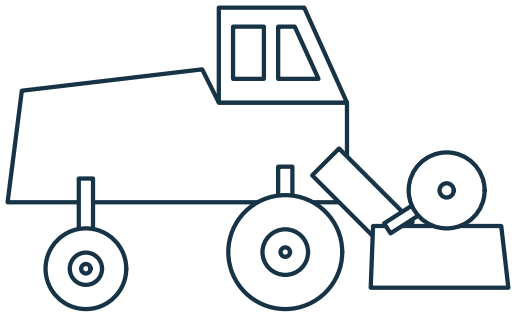
1 Shape the grain body



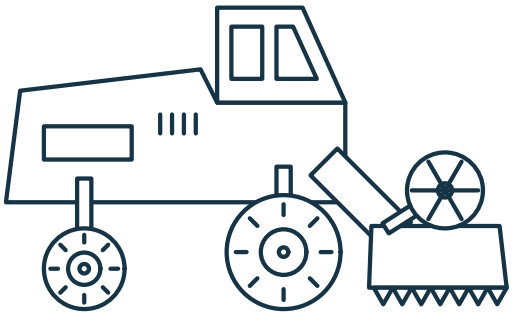
2 Mount cab and both tires



3 Connect header and reel



4 Add reel spokes and teeth



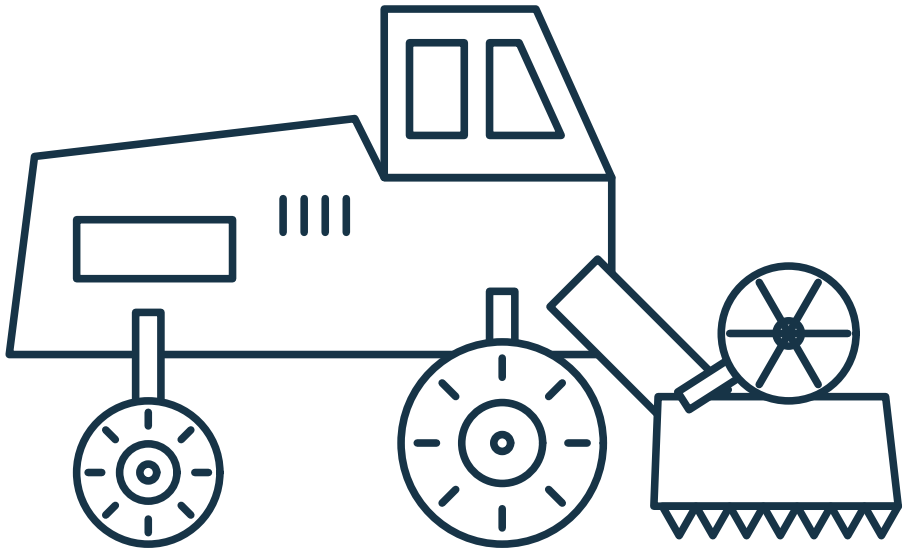
TRY IT

Add rows of wheat in front.

Draw on your own paper. Change one detail to make it yours.

Combine harvester: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Add rows of wheat in front.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 29

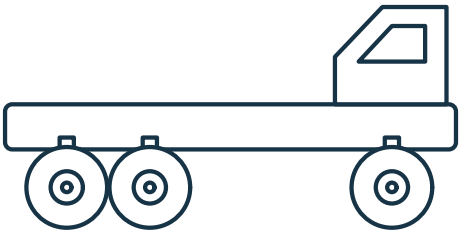
Logging truck

Long trailers carry cut logs.

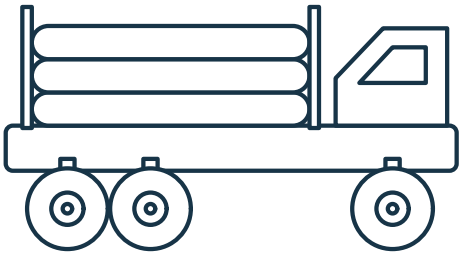
1 Draw the long carrier



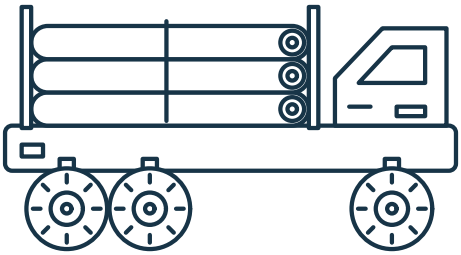
2 Add cab and tires



3 Stack logs inside stakes



4 Add rings and load strap



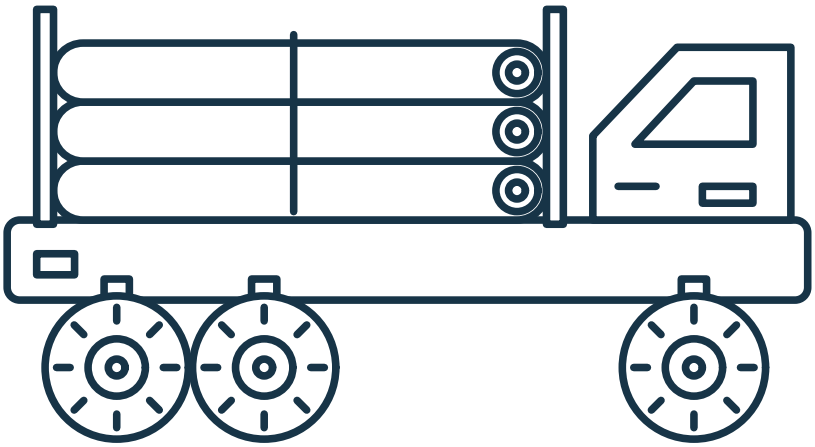
TRY IT

Draw wood rings on the log ends.

Draw on your own paper. Change one detail to make it yours.

Logging truck: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw wood rings on the log ends.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

LEVEL 3 / PROJECT 30

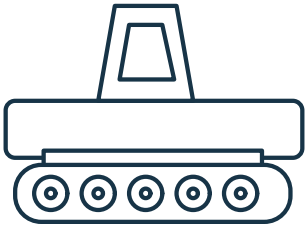
Crawler crane

Tracks support a lattice lifting boom.

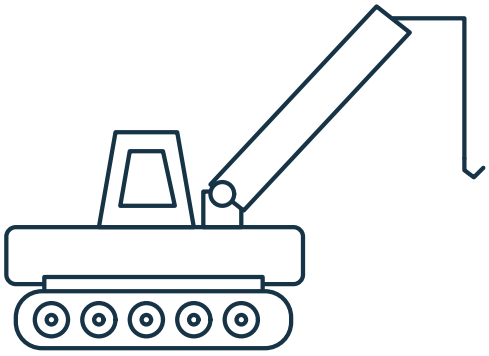
1 Draw the crawler body



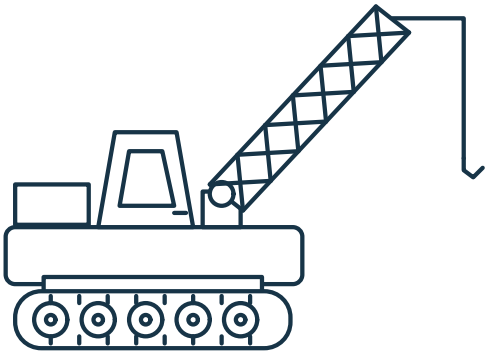
2 Mount tracks and cab



3 Attach boom and hoist



4 Add bracing and counterweight



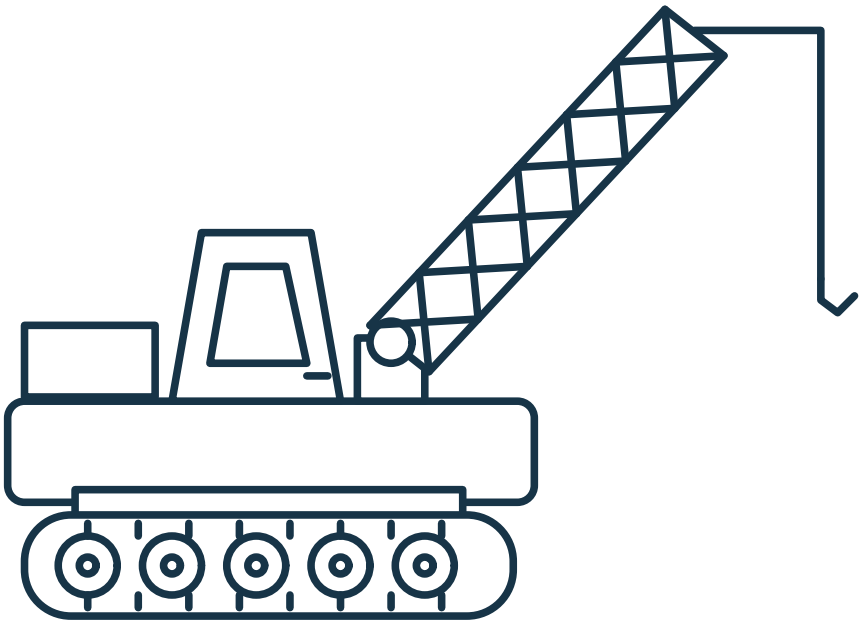
TRY IT

Draw a hook and counterweight.

Draw on your own paper. Change one detail to make it yours.

Crawler crane: draw and color

Study its main parts. Color the outline and add the job-site scene.



Make it your own:

Draw a hook and counterweight.

Circle two moving parts. Add a person for scale.

Use your own paper for a larger version of this machine.

SCENE CHALLENGE 01

A road under repair

Use a paver, roller, and dump truck.

Which machine arrives first? Show the old and new road.

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 02

A new house foundation

Use an excavator, cement mixer, and pump.

Where will the truck stop so its chute can reach?

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 03

A city at sunrise

Use a tower crane, loader, and delivery truck.

Place the crane behind the half-built building.

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 04

A snowy street

Use a snowplow and road grader.

Show where the snow is pushed and where cars may drive.

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 05

A tree cleanup

Use a logging truck, loader, and tow truck.

Keep people away from the moving equipment.

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 06

A harvest day

Use a tractor, combine, and truck.

Show the grain moving from field to truck.

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 07

A tall repair job

Use a scissor lift and boom lift.

Why is each lift useful at a different height?

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 08

A trench for a pipe

Use a trencher, backhoe, and loader.

Make the trench narrow and place the soil beside it.

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 09

A clean neighborhood

Use a street sweeper, fire engine, and forklift.

Make each machine do a different job.

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

SCENE CHALLENGE 10

Invent a job site

Choose any four machines in this book.

Give every machine a task and show what changed.

PLAN THE SCENE

Near: _____ Far: _____

What is each machine doing? _____

Try one version here; make a larger drawing on separate paper.

REFERENCE

The machine finder

01 Dump truck

02 Bulldozer

03 Excavator

04 Wheel loader

05 Road roller

06 Cement mixer

07 Forklift

08 Skid steer

09 Backhoe

10 Mobile crane

11 Tower crane

12 Paver

13 Motor grader

14 Trencher

15 Scraper

16 Telehandler

17 Scissor lift

18 Boom lift

19 Concrete pump

20 Drilling rig

21 Pile driver

22 Asphalt mill

23 Snowplow

24 Street sweeper

25 Tow truck

26 Fire engine

27 Farm tractor

28 Combine harvester

29 Logging truck

30 Crawler crane

NEXT STEPS

Keep drawing

Choose a favorite

Draw it three times: side view, working, and parked.

Change the scale

Draw a tiny worker to show how large the machine is.

Observe in real life

Look for hinges, windows, warning lights, and shadows.

Tell a story

Put two machines in one scene and give each a job.

Remember: start with a shape, then let the details do the work.